SOC Investigation Report #4
File Creation Investigation (Sysmon Event ID 11)

Objective

Detect and investigate file creation events using Sysmon Event ID 11 in Splunk. Determine which process created the file, which user executed the process, and whether the activity is suspicious.

Lab Environment

Component	Details
Host Machine	Windows 11
Virtualization	Oracle VirtualBox
Victim VM	Windows 10 Home
Attacker VM	Kali Linux
SIEM	Splunk Enterprise
Log Forwarder	Splunk Universal Forwarder
Endpoint Monitor	Sysmon v15.21

Commands Used

PowerShell:
mkdir C:\SOC-Lab
New-Item -Path C:\SOC-Lab\TestFile.txt -ItemType File

Verification:
sysmon64 -c

Splunk Queries

index=* EventCode=11

index=* sourcetype="WinEventLog:Microsoft-Windows-Sysmon/Operational" EventCode=11

index=* TargetFilename="*TestFile.txt"

Observed Event

Event ID: 11
Task Category: File created (Rule: FileCreate)
Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
User: WINDOWS\biru
Observed Target File: C:\Users\biru\AppData\Local\Temp__PSScriptPolicyTest_...

Analysis

The event was generated by powershell.exe. The file observed is a temporary PowerShell policy test file created automatically by PowerShell. This is normal operating system behavior and is not malicious.

SOC Conclusion

The investigation identified a legitimate FileCreate event generated by PowerShell. No indicators of compromise were observed. Continue monitoring for unusual file creation in sensitive directories such as Startup, System32, or AppData.

Evidence

Insert screenshots:
1. PowerShell commands creating the folder/file.
2. Splunk Event ID 11 search.
3. Event details highlighting Image, TargetFilename, User and Time.

Skills Learned

• Investigate Sysmon Event ID 11
• Identify the process responsible for file creation
• Interpret TargetFilename and Image fields
• Distinguish legitimate from suspicious file creation activity
• Use Splunk to investigate endpoint events